Alec Schwacha

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SUMMARY

Robotics engineer who has deployed inspection robots at ~100 industrial sites, owned an R&D prototype end to end, and now builds the C++/ROS 2 software that runs them. Designed the Nano prototype's sensor payload, compute unit, and custom EKF on my own; it became the basis for Gecko's next-generation robotic platform. A mechanical engineering background and years in the field shape how I build robots that survive real-world conditions.

EXPERIENCE

Gecko Robotics — Pittsburgh, PA → New York, NY

Jun 2021 – Present

Software Engineer — Aug 2026 – Present

- Built an internal C++ CLI tool that benchmarks localization by replaying bags with known ground-truth trajectories through the EKF and scoring trajectory RMSE, with 3D visualization of the result; wired into CI to fail builds that exceed a 2 cm RMSE threshold
- Built a fiducial tracking system in C++ on ROS 2 Humble that keeps localization robust in feature-poor environments where LiDAR-based methods lose track
- Architected a platform-wide system-health framework that rolls up localization, robot, and system status into real-time go/no-go states, spanning the operator UI down to the backend gate that blocks an inspection when the robot isn't ready
- Implemented the checks that flag degraded conditions — feature-poor environments, sparse point clouds, stale sensors, recording failures — and block invalid customer data collection
- Develop containerized C++ ROS 2 services with Python tooling and Bash environment scripting; validate in Webots simulation before deploying to hardware
- Contribute to the operator front end in Svelte/JavaScript

Forward Deployed Engineer (Hardware) — Feb 2025 – Aug 2026

- Sole engineer on Nano, the R&D prototype for Gecko's expansion into piping inspection, which became the basis for the company's next-generation robotic platform: designed the UT sensor + LiDAR payload and the "Proton Pack" mobile compute unit, end to end
- Implemented a custom EKF in Python on ROS 2 Humble, running on the Proton Pack and fusing LiDAR, IMU, and wheel encoders to publish odometry at 10 Hz, plus hardware-synchronized triggering of the UT sensors
- Designed and built a handheld single-sensor UT probe with 3D localization, bringing position-tagged readings to manual inspections

Test & Reliability Engineer — Mar 2023 – Feb 2025

- Built a Python (Tkinter, pandas) robot telemetry analysis GUI that ingests field telemetry and visualizes robot health, performance trends, and trajectory deviation over time, timestamping when a robot's localization went wrong; replaced manual spreadsheet analysis of Toka fleet performance on job sites
- Built a Python ultrasonic data-quality tool that analyzes A-scans captured by robots, computing SNR per robot per job and thickness distributions to catch degraded sensor data
- Field-tested the Komodo prototype aboard a U.S. Navy aircraft carrier at Pearl Harbor
- Designed test hardware including a UT calibration block, now standard across the Gecko fleet, that removed back-wall reflections rejected by calibration software

Field Services Engineer — Jun 2021 – Mar 2023

- Deployed and operated inspection robots at ~100 customer sites across oil & gas, power generation, pulp & paper, aluminum, waste-to-energy, and U.S. Navy facilities

SKILLS

Languages: C++, Python, Bash, JavaScript, MATLAB
Robotics: ROS 2 (Humble), state estimation (EKF), sensor fusion (LiDAR/IMU/encoders), fiducial tracking, Webots
Tools: Docker, Linux, Git, GitHub Actions, pandas, Tkinter, Svelte
Hardware: Embedded compute integration, sensor payload design, test fixture design, rapid prototyping, machining, GD&T

EDUCATION

University of Pittsburgh — B.S. Mechanical Engineering — 2021

- 2nd place, Swanson School of Engineering Design Expo